

Answers: Mean Class Size

Suppose a college is offering five sections of an introductory statistics course. The class sizes for the sections are:

Section	301	302	303	304	305
Class size	10	4	5	3	6

a) **The 5 sections are the observational units.**

The observational units are the **5 sections**. The variable is the **class size** of each section.

$$\bar{x} = \frac{10 + 4 + 5 + 3 + 6}{5} = \frac{28}{5} = 5.6 \text{ students}$$

b) **The 28 students are the observational units.**

The observational units are the **28 students**. The variable is the **class size of the student's section**. Each student reports the class size of their own section:

Section	Students reporting	Value each reports	Total
301	10	10	100
302	4	4	16
303	5	5	25
304	3	3	9
305	6	6	36
All	28		186

$$\bar{x} = \frac{186}{28} \approx 6.64 \text{ students}$$

The mean is bigger from the students' point of view. More students are in the bigger sections (10 of the 28 are in section 301), so a typical student is in a bigger section than a typical section has. Which mean you get depends on what the observational units are.